Lecture 4

Model Selection, Validation, and Causal Identification

ECON6083: Machine Learning in Economics

Today's Roadmap

#	TOPIC	FOCUS
1	Prediction Foundation	K-Fold CV, hyperparameter tuning
2	Validation Techniques	Metrics for regression/classification
3	Causal Transition	From prediction to causation
4	Causal Identification	CIA, propensity scores, doubly robust
5	LaLonde Case Study	Re-evaluating NSW program

Part 1

The Prediction Foundation

The Fragility of Simple Splits

Why simple train-test splits fail in economics:

- **Heterogeneous data:** Single split highly unstable
- **Sample underutilization:** Costly in small n, large p settings
- **"Luck of the split":** Performance varies with outliers

The problem:

Metrics fluctuate based on which observations go where

K-Fold Cross-Validation

Standard approach for estimating out-of-sample skill:

- Partition data into K equal-sized folds
- Train on $K-1$ folds, test on remaining fold
- Repeat K times, average performance

Heuristic for K:

- Economic practice: $K=5$ or $K=10$
- Balances bias-variance tradeoff
- LOOCV ($K=n$): nearly unbiased but high variance

Stratified K-Fold

Essential for imbalanced outcomes

When to use:

- Credit scoring with rare defaults
- Policy targeting with low treatment prevalence
- Any classification with rare positive class

Key benefit:

Maintains original class distribution across all folds

Without stratification:

- Some folds may have zero positive cases
- Model cannot learn the minority class pattern
- Fold-level metrics become meaningless

K-Fold CV in Python

```python
from sklearn.model_selection import cross_val_score, KFold
from sklearn.linear_model import Ridge
import numpy as np

kf = KFold(n_splits=5, shuffle=True, random_state=42)
model = Ridge(alpha=1.0)

# Returns one score per fold
scores = cross_val_score(
    model, X, y, cv=kf,
    scoring="neg_mean_squared_error"
)
rmse = np.sqrt(-scores)
print(f"Mean RMSE: {rmse.mean():.3f} ± {rmse.std():.3f}")
```

Hyperparameter Tuning

Method	Exhaustive grid	Random sampling
Speed	Slow (exponential)	Fast (fixed budget)
Best for	Low-dim (1-2 params)	High-dim (5+ params)
Trade-off	Thorough but expensive	Efficient, may miss optimum

Examples:

- **Ridge** (λ only): Grid Search — 6 values suffice
- **Random Forest** (4 params): Random Search — 5^4=625 too many
- **XGBoost** (5+ params): Random Search or Bayesian

GridSearchCV in Python

```
from sklearn.model_selection import GridSearchCV
from sklearn.ensemble import RandomForestRegressor

param_grid = {
    "n_estimators": [100, 300, 500],
    "max_depth": [5, 10, 20],
    "min_samples_leaf": [5, 10, 20],
}
grid = GridSearchCV(
    RandomForestRegressor(random_state=42),
    param_grid, cv=5,
    scoring="neg_mean_squared_error",
    n_jobs=-1,
)
grid.fit(X_train, y_train)
print(f"Best params: {grid.best_params_}")
print(f"Best RMSE: {(-grid.best_score_)**0.5:.3f}")
```

Pipeline: Preventing Data Leakage

```
from sklearn.pipeline import Pipeline
from sklearn.preprocessing import StandardScaler
from sklearn.linear_model import LassoCV

# Scaling fit ONLY on training folds — no leakage
pipe = Pipeline([
    ("scaler", StandardScaler()),
    ("lasso", LassoCV(cv=5)),
])

scores = cross_val_score(
    pipe, X, y, cv=5,
    scoring="neg_mean_squared_error"
)
print(f"Mean RMSE: {(-scores.mean())**0.5:.3f}")
```

Key: GridSearchCV finds best params; Pipeline prevents leakage. Use both.

Bootstrap Confidence Intervals

Problem: CV gives a point estimate of performance; we need uncertainty.

Bootstrap approach:

1. Draw B resamples (with replacement) from the data
2. For each resample, compute CV error
3. Use percentiles of the B errors as CI

$$CI_{0.95} = [\hat{\theta}^*_{0.025}, \hat{\theta}^*_{0.975}]$$

Why bootstrap for CV:

- Accounts for both sampling variation AND split variation
- More honest than standard CV SE formulas

Reference: Efron and Tibshirani (1993), *An Introduction to the Bootstrap*.

Time Series Cross-Validation

Problem: Standard K-fold shuffles data → violates temporal ordering

Expanding window CV:

```
Train: [1]      Test: [2]
Train: [1,2]    Test: [3]
Train: [1,2,3] Test: [4]
...
```

Blocked CV: Preserve local dependence structure; block size = autocorrelation length

When to use:

- Forecasting (macro, finance)
- Policy evaluation with rolling windows
- Any data with serial correlation

Bias-Variance Decomposition

Expected prediction error at point x:

$$E[(Y - \hat{f}(x))^2] = \underbrace{\text{Bias}^2[\hat{f}(x)]}_{\text{Systematic error}} + \underbrace{\text{Var}[\hat{f}(x)]}_{\text{Sensitivity to data}} + \underbrace{\sigma_\varepsilon^2}_{\text{Irreducible noise}}$$

MODEL	BIAS	VARIANCE	TYPICAL EXAMPLE
Too simple	High	Low	OLS with few features
Too complex	Low	High	Deep neural network, small n
Just right	Balanced	Balanced	Random Forest, tuned Lasso

> *Cross-validation estimates the **total error**; diagnostic tools (learning curves) separate bias vs variance.*

Permutation Feature Importance

Question: Which features drive model performance?

Method:

1. Train model, record baseline CV score
2. For each feature j: randomly shuffle its values (breaking $Y \sim X_j$ relationship)
3. Recompute CV score
4. Importance = baseline – shuffled score

Advantage: Model-agnostic — works for any black-box model

Caution: Correlated features share importance; individual importances can be misleading

Lasso λ Selection

Cross-validation approach:

- Standard practice in ML
- Computationally intensive

Rigorous Lasso (Belloni & Chernozhukov 2013):

- Plug-in penalty based on noise level
- Avoids heavy CV cost
- Stronger theoretical guarantees for inference

Part 2

Evaluation Metrics

The Confusion Matrix

	PREDICTED 0	PREDICTED 1
Actual 0	True Negative (TN)	False Positive (FP)
Actual 1	False Negative (FN)	True Positive (TP)

Four possible outcomes for each prediction

Classification Metrics

Accuracy: $(TP + TN)/N$

- Misleading for imbalanced data!

Precision: $TP/(TP + FP)$

- "If model says default, is it correct?"

Recall (Sensitivity): $TP/(TP + FN)$

- "Did we catch all the defaults?"

F1 Score: Harmonic mean of precision and recall

ROC Curve and AUC

ROC (Receiver Operating Characteristic):

- Plots True Positive Rate vs False Positive Rate
- Across all classification thresholds

AUC (Area Under Curve):

- Single number summary
- AUC = 0.5: Random guessing
- AUC = 1.0: Perfect classifier
- Robust to class imbalance

The accuracy trap: With 1% positive rate, predicting "always negative" gives 99% accuracy but AUC = 0.5.

SHAP Values: Explaining Predictions

Problem: Random Forest/XGBoost are "black boxes"

SHAP (SHapley Additive exPlanations):

- Decomposes each prediction into feature contributions
- Answers: "Why did model predict default for this person?"

LEVEL	QUESTION	USE CASE
Global	Which features matter most?	Model validation
Local	Why this specific prediction?	Loan denials

Key insight: Even black box models can be explained.

Part 3

Causal Identification Frameworks

Prediction vs Causation

Question	"Who will default?"	"Does raising rate *cause* defaults?"
Data	Income, credit score	Same + exogenous variation
Action	Deny high-risk loans	Set optimal interest rate
Key	Correlation sufficient	Need to isolate causal effect

Why it matters: Good prediction $\neq$ Good policy

Potential Outcomes Framework

Fundamental setup:

- $Y(1)$: outcome under treatment
- $Y(0)$: outcome under control
- Only one observed per individual

The fundamental problem:

Cannot observe both $Y(1)$ and $Y(0)$ for same person

Causal effect:

$\tau_i = Y_i(1) - Y_i(0)$ (individual level, unobserved)

Conditional Independence Assumption (CIA)

The Problem: People choose treatment for a reason

- College-goers differ from non-goers (ability, motivation)
- Can't compare their incomes directly

CIA Solution: Compare people who look similar on observables X

Assumption: Within this group, college choice is "as good as random"

When CIA fails:

- Unmeasured ability affects both enrollment and earnings
- Remaining difference is selection bias, not causal effect

Overlap (Positivity)

The Problem: Need comparable units

- Can we estimate effect of MBA for 18-year-olds?
    - No: $P(\mathrm{MBA} = 1 \mid \mathrm{age} = 18) \approx 0$ — no counterfactual
- Can we estimate effect for 30-year-olds with bachelor's?
    - Yes: $0 < P(\mathrm{MBA} = 1) < 1$ — some have MBA, some don't

Rule of thumb: Trim observations with $p(X) < 0.05$ or > 0.95

Checking Covariate Balance

After matching or weighting: verify that X distributions are similar across treatment groups

Standardized mean difference (SMD):

$$\text{SMD}_j = \frac{\bar{X}_{j,treat} - \bar{X}_{j,ctrl}}{\sqrt{(s^2_{j,treat} + s^2_{j,ctrl})/2}}$$

Rule of thumb: SMD < 0.1 for all covariates → acceptable balance

Common mistake: Only check balance on means. Also check variance ratios and Kolmogorov-Smirnov statistics.

Propensity Score Matching

Propensity score:

$$p(X) = P(D = 1 \mid X)$$

Key property (Rosenbaum & Rubin 1983):
Balancing score — condenses high-dimensional X into scalar

$$(Y(0), Y(1)) \perp D \mid p(X)$$

Strategy: Pair treated and control units with similar $p(X)$

Common approach: 1:1 nearest-neighbor matching with caliper

Inverse Propensity Weighting (IPW)

ATE estimator:

$$\hat{\tau}_{IPW} = \frac{1}{n} \sum_{i=1}^{n} \left[\frac{D_i Y_i}{\hat{p}(X_i)} - \frac{(1 - D_i) Y_i}{1 - \hat{p}(X_i)} \right]$$

Intuition: Weight observations by inverse treatment probability to create a pseudo-population where treatment is independent of X *(illustrative numbers)*

PERSON	PROB	WEIGHT	EFFECT
Rich + Treated	70%	$1/0.7 = 1.43$	Down-weighted
Poor + Treated	30%	$1/0.3 = 3.33$	Up-weighted

Doubly Robust (AIPW) Estimator

Augmented IPW:

$$\hat{\tau}_{DR} = \frac{1}{n}\sum_{i=1}^{n}\left[\frac{D_i(Y_i - \hat{m}_1(X_i))}{\hat{p}(X_i)} - \frac{(1-D_i)(Y_i - \hat{m}_0(X_i))}{1-\hat{p}(X_i)} + \hat{m}_1(X_i) - \hat{m}_0(X_i)\right]$$

Doubly robust property: Consistent if **either** $\hat{p}$ or $\hat{m}$ is correct

$\hat{p}(X)$	$\hat{m}(X)$	
Correct	Wrong	Consistent
Wrong	Correct	Consistent
Both correct	Both correct	Most efficient

Part 4

The LaLonde Case Study

Paper 1: LaLonde (1986, AER)

"Evaluating the Econometric Evaluations of Training Programs with Experimental Data"
American Economic Review, 1986

Research question: Can standard econometric methods recover experimental estimates of job training effects?

ML: OLS, fixed effects, matching — benchmark paper showing these fail without good comparison groups

The test: Replace experimental controls with CPS/PSID survey data; do methods recover the experimental benchmark?

LaLonde (1986): Findings

METHOD	ESTIMATE	VS BENCHMARK	PROBLEM
Experimental	$886	Benchmark	True effect
OLS (simple)	$-1,556	Wrong sign	Selection bias
OLS (full controls)	$628	30% bias	Model dependence
Fixed Effects	Varies widely	Unstable	Comp. group

Source: LaLonde (1986, Table 3; experimental benchmark $886). Non-experimental estimates from CPS/PSID comparison groups.

Why methods failed:

1. Selection bias: trainees differ systematically from population
2. Comparison group mismatch: CPS respondents not comparable
3. Estimates highly sensitive to specification

Paper 2: Dehejia & Wahba (1999, JASA)

"Causal Effects in Nonexperimental Studies: Reevaluating the Evaluation of Training Programs: Reevaluating the Evaluation of Training Programs"
Journal of the American Statistical Association, 1999

Research question: Can propensity score matching recover the experimental benchmark with a better comparison group?

ML: Propensity score matching (logistic regression) with nearest-neighbor, radius, and kernel matching

FACTOR	LALONDE	DEHEJIA & WAHBA
Comparison group	Full CPS/PSID	Restricted CPS subsample
Overlap	Poor	Good
Result	Methods failed	PSM succeeded

Dehejia & Wahba (1999): Results

SPECIFICATION	ESTIMATE	SE
Experimental	$1,794	$671
PSM (Radius=0.1)	$1,672	$937
PSM (Radius=0.05)	$1,611	$979
Linear regression	$1,550	$838

Source: Dehejia & Wahba (1999, JASA, Table 1).

Success factors:

1. Better comparison group — restricted CPS more similar to trainees
2. Overlap enforcement — PSM drops poorly matched observations
3. Covariate balance ensured

Lesson: Data quality > method choice

Paper 3: Angrist & Krueger (1991, QJE)

"Does Compulsory School Attendance Affect Schooling and Earnings?"
Quarterly Journal of Economics, 1991

Research question: What is the causal return to an additional year of education?

ML: Instrumental variable (birth quarter) — no ML used; DML revisits this setting in Lecture 8

MECHANISM	EXPLANATION
School entry laws	Children born Jan–Dec enter same grade
Dropout behavior	Students can leave at age 16
Result	Born earlier → age 16 in earlier grade → less schooling

Angrist & Krueger (1991): Results

First Stage: Born Q4 vs Q1 → approximately +0.1 years of education

SPECIFICATION	RETURN TO EDUCATION
OLS	7.1%
IV (birth quarter)	7.5%

Source: Angrist & Krueger (1991, QJE, Table IV, Column 2 for IV). Based on 1980 U.S. Census, men born 1930–1939.

Interpretation:

- OLS not severely biased (contrary to early critics)
- IV identifies LATE for "compliers" (induced to stay by school laws)
- Birth quarter affects schooling but not earnings directly (exclusion restriction)

Three Papers: Common Themes

	LALONDE (1986)	DEHEJIA & WAHBA (1999)	ANGRIST & KRUEGER (1991)
Core Issue	Selection bias	Comparison group quality	Endogeneity
Solution	Experimental benchmark	Better matching + overlap	Exogenous variation
Key Insight	Non-exp methods can fail	Data quality > method	Creative IVs

Bottom Line: Credible causal inference requires an identification strategy — ML helps but doesn't replace careful design

Common Pitfalls

1. Data leakage in CV

- Preprocessing on entire dataset before CV leaks test info into training

2. Using accuracy with imbalanced data

- 99% accuracy with 1% positive rate is meaningless
- Use Precision, Recall, F1, AUC instead

3. Ignoring sample size in K-fold

- Small n: use leave-one-out or $K = 5$
- Large n: $K = 10$ is fine

4. Time series without forward chaining

- Standard CV violates temporal structure
- Use expanding window or blocked CV

Key Takeaways

TOPIC	KEY POINT
K-Fold CV	Stratified K-fold ($K = 5$ or 10); forward-chaining for time series
Tuning	Grid (few params) vs Random (many params)
Metrics	Don't trust accuracy with imbalance; use Precision/Recall/AUC
CIA	Conditioning on X makes treatment "as good as random"
Overlap	Trim extreme propensity scores for credibility
AIPW	Doubly robust: consistent if either model correct

Further Reading

PAPER	AUTHORS	VENUE
Evaluating the Econometric Evaluations of Training Programs w/ Experimental Data	LaLonde (1986)	*AER*
Causal Effects in Nonexperimental Studies: Reevaluating the Evaluation of Training Programs	Dehejia & Wahba (1999)	*JASA*
Does Compulsory School Attendance Affect Schooling and Earnings?	Angrist & Krueger (1991)	*QJE*
An Introduction to the Bootstrap	Efron & Tibshirani (1993)	Chapman & Hall
Inference on Treatment Effects after Selection among High-Dimensional Controls	Belloni, Chernozhukov & Hansen (2014)	*ReStud*

Data and Code

This lecture:

- Slides source: `_slides/Lecture04-Cross-Validation and Model Selection-Slides.md`
- Exercises: `exercises/Lecture04-Exercise.ipynb`

Key packages:

- Python: `scikit-learn` (GridSearchCV, RandomizedSearchCV)

Datasets:

- Simulated DGPs; LaLonde NSW dataset

Thank You!

Next Lecture: Double/Debiased Machine Learning

Reading: Chernozhukov et al. (2018), "Double/Debiased ML"